The Derivative from First Principles

ANSWER KEY

The definition of the derivative

1 Use the limit definition $f'(x) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$ to find $f'(x)$ for $f(x) = x^2 + 3x$.
$\frac{(x+h)^2 + 3(x+h) - (x^2+3x)}{h} = \frac{2xh + h^2 + 3h}{h} = 2x + h + 3 \to 2x + 3$.

2 Use the limit definition to find $f'(x)$ for $f(x) = \sqrt{x}$ (for $x > 0$).
$\frac{\sqrt{x+h} - \sqrt{x}}{h} = \frac{h}{h(\sqrt{x+h} + \sqrt{x})} = \frac{1}{\sqrt{x+h} + \sqrt{x}} \to \frac{1}{2\sqrt{x}}$.

3 The graph shows $f(x) = x^2$ with the secant line through $(1, 1)$ and $(3, 9)$.

a Find the slope of the secant line. $\frac{9-1}{3-1} = 4$.

b Find $f'(2)$ using the derivative, and compare it to the secant slope. $f'(x) = 2x$, so $f'(2) = 4$ — for this quadratic, the secant slope between $x = 1$ and $x = 3$ happens to equal the instantaneous slope at the midpoint $x = 2$.

Differentiability

4 Explain why $f(x) = |x|$ is not differentiable at $x = 0$, using one-sided derivatives.
The left-hand derivative is $\lim_{h\to 0^-}\frac{|h|}{h} = -1$ and the right-hand derivative is $\lim_{h\to 0^+}\frac{|h|}{h} = 1$. Since these disagree, $f'(0)$ does not exist.

5 State the relationship between differentiability and continuity: if a function is differentiable at a point, must it be continuous there? Is the converse true?
Differentiability implies continuity (a corner or asymptote would break the limit needed for the derivative). The converse is false — $f(x) = |x|$ is continuous at 0 but not differentiable there.

More derivatives from the definition

6 Use the limit definition to find $f'(x)$ for $f(x) = \frac{1}{x}$ (for $x \neq 0$).
$\frac{\frac{1}{x+h} - \frac{1}{x}}{h} = \frac{x - (x+h)}{hx(x+h)} = \frac{-h}{hx(x+h)} = \frac{-1}{x(x+h)} \to -\frac{1}{x^2}$.

7 Use the limit definition to find $f'(x)$ for $f(x) = x^3$.
$(x+h)^3 = x^3 + 3x^2h + 3xh^2 + h^3$, so $\frac{f(x+h) - f(x)}{h} = \frac{3x^2h + 3xh^2 + h^3}{h} = 3x^2 + 3xh + h^2 \to 3x^2$.

Interpreting the derivative

8 A ball's height is $h(t) = -5t^2 + 20t + 2$ (metres, seconds). Use the derivative definition idea to explain what $h'(2)$ represents physically, and find its value using $h'(t) = -10t + 20$.

$h'(2)$ represents the instantaneous vertical velocity of the ball at $t = 2$ seconds.
$h'(2) = -10(2) + 20 = 0$ m/s — the ball is momentarily at rest, at its peak height.

Point-derivatives from the definition

9 Use the limit definition $f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$ to find $f'(4)$ for $f(x) = x^2 - 3x$.

$f(4) = 4$. $f(4+h) = (4+h)^2 - 3(4+h) = 16 + 8h + h^2 - 12 - 3h = 4 + 5h + h^2$.
$\frac{f(4+h) - f(4)}{h} = \frac{5h + h^2}{h} = 5 + h \to 5$: $f'(4) = 5$.

10 Use the alternate form $f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$ to find $f'(2)$ for $f(x) = x^3 - 1$.

$f(2) = 7$. $\frac{x^3 - 1 - 7}{x - 2} = \frac{x^3 - 8}{x - 2} = \frac{(x-2)(x^2 + 2x + 4)}{x - 2} = x^2 + 2x + 4 \to 4 + 4 + 4 = 12$: $f'(2) = 12$.

The derivative as the slope of the tangent line

11 For $f(x) = x^2 - 4x + 1$, use the derivative (found via the limit definition, $f'(x) = 2x - 4$) to find the equation of the tangent line at $x = 3$.

$f(3) = 9 - 12 + 1 = -2$. $f'(3) = 2(3) - 4 = 2$. Tangent line: $y - (-2) = 2(x - 3)$, i.e. $y = 2x - 8$.

12 Explain why the tangent line to $y = x^2$ at $x = 0$ is horizontal, using the derivative $f'(x) = 2x$ found from the limit definition.

$f'(0) = 2(0) = 0$. Since the slope of the tangent line at $x = 0$ is 0, the tangent line is horizontal there — consistent with $x = 0$ being the vertex (minimum point) of the parabola.

The derivative function versus the derivative at a point

13 Explain the difference between $f'(x)$ and $f'(3)$, using $f(x) = x^2$ as an example.

$f'(x) = 2x$ is a FUNCTION giving the slope of the tangent line at any point x. $f'(3) = 2(3) = 6$ is a single NUMBER — the specific slope at the point $x = 3$. The function $f'(x)$ can be evaluated at any x to get such a number.

Estimating a derivative from a table

14 A table gives $f(1.9) = 8.61$, $f(2.0) = 9.00$, $f(2.1) = 9.41$. Use a symmetric difference quotient to estimate $f'(2)$.

$f'(2) \approx \frac{f(2.1) - f(1.9)}{2.1 - 1.9} = \frac{9.41 - 8.61}{0.2} = \frac{0.80}{0.2} = 4$.

The derivative of a constant and linear functions

15 Use the limit definition to show that the derivative of a constant function $f(x) = c$ is 0.

$\frac{f(x+h)-f(x)}{h} = \frac{c-c}{h} = \frac{0}{h} = 0$ for all $h \neq 0$, so $f'(x) = \lim_{h\to 0} 0 = 0$.

16 Use the limit definition to show that the derivative of a linear function $f(x) = mx + b$ is the constant m.

$\frac{f(x+h)-f(x)}{h} = \frac{m(x+h)+b-(mx+b)}{h} = \frac{mh}{h} = m$ for all $h \neq 0$, so $f'(x) = \lim_{h\to 0} m = m$.